

Multiple Choice (10 marks)

1. A and B are two events. If $P(A, B)$ decreases while $P(A)$ increases, which of the following is true?
 - (a) $P(A|B)$ decreases
 - (b) $P(B|A)$ decreases
 - (c) $P(B)$ decreases
 - (d) All of above
2. Statement 1| Industrial-scale neural networks are normally trained on CPUs, not GPUs.
Statement 2| The ResNet-50 model has over 1 billion parameters.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. Consider the Bayesian network given below. How many independent parameters are needed for this Bayesian Network $H \rightarrow U \leftarrow P \leftarrow W$?
 - (a) 2
 - (b) 4
 - (c) 8
 - (d) 16
4. Statement 1| The softmax function is commonly used in multiclass logistic regression.
Statement 2| The temperature of a nonuniform softmax distribution affects its entropy.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
5. Which one of the following is the main reason for pruning a Decision Tree?
 - (a) To save computing time during testing
 - (b) To save space for storing the Decision Tree
 - (c) To make the training set error smaller
 - (d) To avoid overfitting the training set

True / False (10 marks)

Answer True or False

6. True or False: Random crop and horizontal flip are among the most common image data augmentation techniques used for natural images.
7. True or False: Overfitting is more likely when the set of training data is small, as a larger dataset typically improves generalization.
8. True or False: To calculate $P(H|E, F)$, the sufficient numbers are $P(E, F)$, $P(H)$, and $P(E, F|H)$.
9. True or False: Discriminative approaches aim to model the joint probability $p(y, x)$, capturing both input features and output labels simultaneously.
10. True or False: The tanh function is a linear activation function commonly used in neural networks.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of an increasing training loss with the number of epochs in a machine learning model. What potential issues could arise, particularly regarding the choice of step size?
12. Discuss the joint probability of the variables H , U , P , and W in the specified Bayesian Network $H \rightarrow U \leftarrow P \leftarrow W$, detailing how to derive it from the conditional probabilities involved.

End of Paper